cost of the hardware involved.

4. The use and implementation of emerging technologies, such as edge computing, is still very expensive.

5. Large corporations (from India) have made lesser investment in AI electronics hardware as compared to startups and global corporations.

6. There are no courses in colleges and universities that focus entirely on AI and ML.

7. AI and ML need to be given a solution-centric approach rather than a product-centric approach.

8. Aggregator platforms are missing.

9. We do not have local R&D centres to create global products.

"We have just missed the bus in IoT excitement. IoT was a big hype in the 2013-14 timeframe. What's happening in 2020 in India is a very low-value addition in terms of system integration. Let's not make the same mistake in the AI ecosystem. Thinking of the hardware side, there is a whole bunch of unsolved problems that are both challenges as well as opportunities. We need to become the value creators for the world," explained Ashok Mishra.

Opportunities

1. AI and emerging technologies electronics hardware is a very big opportunity for entrepreneurs. We need to develop applications that work on economical hardware.

2. India's specific solutions will be in huge demand. These are further going to encourage the adoption of AI and data science.

3. Indigenous chip manufacturing and development in the domestic sector will grow in India. This segment will see a good jump in the AI hardware sector.

4. Individuals who can give guest lectures around emerging technologies will be needed. More faculties will be required.

5. India can emerge as the leader of the world in terms of design houses.

6. New labs, academia, and testing centres for electronics hardware around AI and emerging technologies will be required.

7. Sectors ranging from automation and robotics to agriculture and manufacturing will see a big boost in the adoption of AI and emerging tech.

8. There is a big scope for multidisciplinary smart companies to emerge as the opportunity is very huge.

"The hardware requirement for AI, ML, deep learning, and other emerging technologies is growing at a phenomenal pace. The amount of computing is doubling at every three and a half months. This is the fastest growth ever. We need to think out of the box as the traditional methods won't suffice. This is where the opportunity and challenges lie for everyone in the domain," said Bharat Daga.

What can be done?

1. We need to have a cohort of a pool of skilled people in these technologies. Academia can play a big role in the same by training individuals and budding engineers.

2. Collaboration between startups, academia, and corporate houses can work wonders as it helps approach the solution from different angles.

3. The government needs to be the catalyst for adoption. It should be open to collaborate with the ecosystem working on the front.

4. Engineers, especially from the hardware and mechanical background, need to learn more about AI. They can upskill, grow, and contribute more.

5. Around 1700 startups are focused on AI. The advantage of being a startup is that it is small-sized and can experiment.

6. A fine balance between theory and practicals should be introduced. While classes can be taught online by the industry, practicals must be hands-on in a good lab setup.

7. Corporates and startups need to take a deep dive into the AI hardware space.

8. AI and ML are horizontal. We need to stop fitting these into other courses and give them a vertical approach.

9. We need to develop a collaborative culture to act as a catalyst. The government needs to play a big role in the same.

"If we look back at some of the startling things that have happened, we may find that the government plays a crucial role. Any massive transformation has its roots in the government. It is wonderful to see ESSCI saying that we need to bring all the stakeholders together. It has to be a collaborative effort from startups, corporates, academia, and the government," concluded Sathya Prasad. EFY